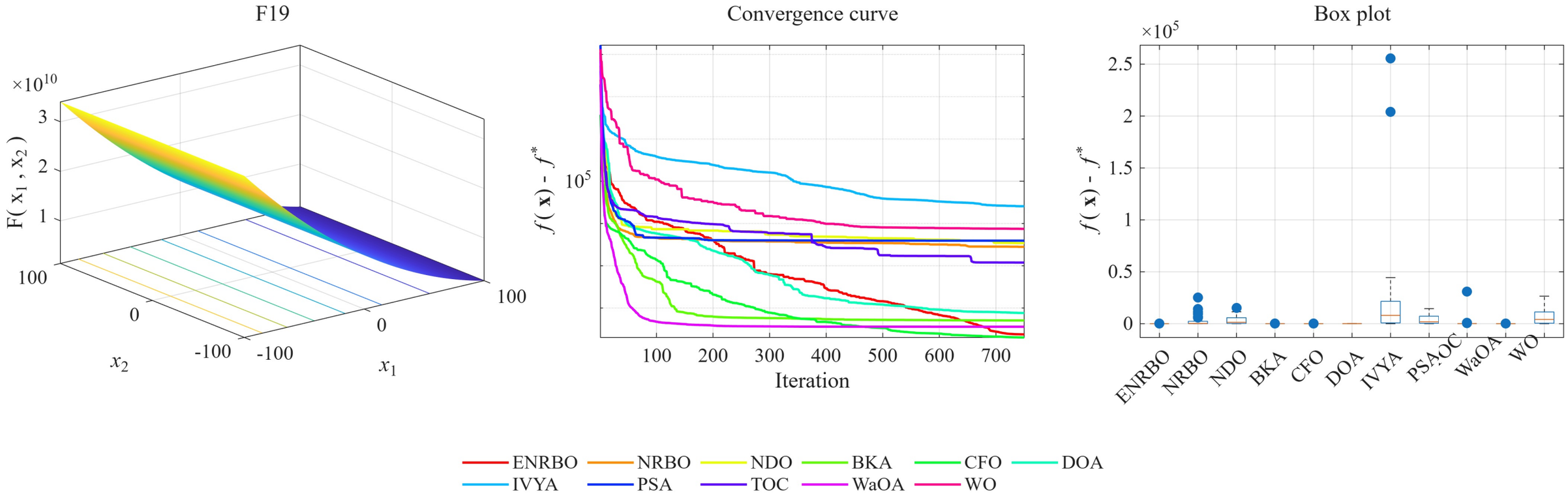

F19
×10^10
3
2
1
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
10^5
100
200
300
400
500
600
700
Iteration
Box plot
×10^5
2.5
2
1.5
1
0.5
0
f(x) - f*
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO